

Ultra-high-frequency ultrasound and machine learning approaches for the differential diagnosis of melanocytic lesions

Francesco Faita¹ | Teresa Oranges^{2,3}  | Nicole Di Lascio¹ | Francesco Ciompi⁴ | Saverio Vitali⁵ | Giacomo Aringhieri⁵ | Agata Janowska² | Marco Romanelli² | Valentina Dini²

¹Institute of Clinical Physiology, National Research Council, Pisa, Italy

²Department of Dermatology, University of Pisa, Pisa, Italy

³Dermatology Unit, Department of Pediatrics, Meyer Children's University Hospital, Florence, Italy

⁴Computational Pathology Group, Radboud University Medical Center, Nijmegen, The Netherlands

⁵Diagnostic and Interventional Radiology, University of Pisa, Pisa, Italy

Correspondence

Teresa Oranges, Department of Dermatology, University of Pisa, Via Roma 67, 56126, Pisa, Italy.
Email: teresa.oranges@gmail.com

Abstract

Malignant melanoma (MM) is one of the most dangerous skin cancers. The aim of this study was to present a potential new method for the differential diagnosis of MM from melanocytic naevi (MN). We examined 20 MM and 19 MN with a new ultra-high-frequency ultrasound (UHFUS) equipped with a 70 MHz linear probe. Ultrasonographic images were processed for calculating 8 morphological parameters (area, perimeter, circularity, area ratio, standard deviation of normalized radial range, roughness index, overlap ratio and normalized residual mean square value) and 122 texture parameters. Colour Doppler images were used to evaluate the vascularization. Features reduction was implemented by means of principal component analysis (PCA), and 23 classification algorithms were tested on the reduced features using histological response as ground-truth. Best results were obtained using only the first component of the PCA and the weighted k-nearest neighbour classifier; this combination led to an accuracy of 76.9%, area under the ROC curve of 83%, sensitivity of 84% and specificity of 70%. The histological analysis still remains the gold-standard, but the UHFUS images processing using a machine learning approach could represent a new non-invasive approach.

KEYWORDS

high-frequency ultrasound, machine learning, melanocytic naevi, melanoma, ultrasonography

1 | BACKGROUND

Malignant melanoma (MM) is a melanocytic tumor characterized by high morbidity and mortality.¹ Advanced MM is associated with poor long-term survival and early MM detection improves prognosis and survival rates.² Differentiating between MM and atypical melanocytic nevi (MN) can often be difficult; therefore, additional non-invasive methods could assist with diagnostic accuracy.

2 | PREMISES

Sensitivity and specificity values in MM diagnosis on visual inspection have been reported to be highly variable with values ranging 67%–86% and 77%–79%, respectively.^{3,4} Dermoscopy may help in performing early-stage detection of MM, with reported sensitivity and specificity values in the MM diagnosis of 91% and 74%.^{3–6} However, it is not recommended for untrained users.⁷ In vivo reflectance confocal microscopy is another useful technique in MM

diagnosis.⁸ Historically, high-frequency ultrasound (20 MHz) has been used for the analysis of skin tumor thickness.^{9–11} Despite the limitations of sonography to detect pigments, the assessment of morphological and vascular characteristics of the lesion may contribute to the formulation of the diagnosis.¹¹ Modern dermatologic ultrasound systems are characterized by a spatial resolution equal to 80 microns. However, only few works discussed the adoption of ultrasound imaging for the characterization and differentiation between MM and MN.^{12,13} Artificial intelligence and machine learning applications in the field of medical image classification are rapidly growing.^{14,15} This is allowing a wider diffusion of computer-aided diagnosis systems, which can be used to improve differential diagnosis.

3 | HYPOTHESIS

The use of a new ultra-high-frequency ultrasound (UHFUS) system (Vevo[®]MD, Fujifilm, Visualsonics), equipped with a 70 MHz linear probe, may help differentiate between MM and MN. This new device can scan lesions to the depth of 1 cm providing a spatial resolution of 30 microns, allowing to obtain high-quality ultrasound images even in the case of small lesions. The use of a machine learning approach in processing these ultrasonographic images could be useful to discriminate between MM and MN. Features obtained with the automatic analysis of the images can be used to train and validate classifiers based on machine learning algorithms and help in the automated differentiation of MM from MN based on the use of UHFUS data.

3.1 | How to test the hypothesis

To verify the hypothesis, it is necessary to evaluate MM and MN with UHFUS (70 MHz) before the excisions, with both brightness mode (B-mode) and colour Doppler acquisitions. A machine learning approach can be used in processing ultrasonographic images to amplify the possibility to differentiate between MM and MN.

4 | PRELIMINARY SUPPORTING EVIDENCE

4.1 | Methods

4.1.1 | Subjects

We analysed 39 melanocytic lesions (20 MM and 19 MN) selected for atypical features during dermatologic visits at the Department of Dermatology (University of Pisa, Pisa, Italy) from July 2016 to March 2017. The study population consisted of 20 caucasian patients with MM (7 women and 13 men), aged from 26 to 84 years (mean age 56.8) and 19 caucasian patients with MN (12 women and 7 men), aged from 15 and 60 years (mean age 37.6). Clinical, dermatoscopic

and ultrasonographic evaluations were performed in all patients. Informed consent and the approval of the local research ethics committee were obtained. The lesions were removed, with margins of excision according to the European guidelines¹⁶ in case of clinical MM suspicion, and histologically examined.

4.1.2 | Ultrasound acquisition

Ultrasonographic data were acquired during clinical examination by two experienced dermatologists before excision using a new UHFUS (Vevo[®]MD, Fujifilm, Visualsonics). For the acquisitions, a 70 MHz probe (HF-70, Visualsonics, SonoSite) was employed. For all the lesions, both B-mode and colour Doppler clips were recorded. B-mode acquisitions of the entire lesion were obtained in the longitudinal view. Colour Doppler images were achieved from the same scan projection using a pulse repetition frequency (PRF) of 5 kHz. A single image was selected from each B-mode clip as that presenting the deepest section of the lesion; this image was used for the further processing. For each image, the lesion was segmented by manually identifying its edges. The located edge points were employed to evaluate following 8 morphological features: area, perimeter, circularity, area ratio, standard deviation of normalized radial range, roughness index, overlap ratio and normalized residual mean square value. Once the tumor edge was detected, texture analysis was applied on the pixels belonging to the lesion; this analysis led to the calculation of 122 texture features. Both the morphological and the texture parameters are summarized in Table 1. A single image was obtained from each of the colour Doppler clips with the minimum noise related to movements. For each image, the presence of intralesional and sublesional vascularization was recorded. Moreover, this image was used for further processing, thus obtaining 2 vascularization parameters (Table 1). For each image, a region of interest corresponding to the lesion was manually identified. All the post-processing operations were performed by an operator with experience in image processing using customized software developed using MATLAB R2016b (MathWorks Inc.). Placement of the region of interest was approved by an expert dermatologist.

4.1.3 | Features selection

Features reduction was applied before the classification procedure. For this purpose, principal component analysis (PCA) was employed to reduce the dimensionality of the predictor space.¹⁷ In the present study, only PCA components explaining the 95% of the variance were kept to feed the model.

4.1.4 | Classification

Different classification models were trained using the MATLAB Statistics and Machine Learning Toolbox (MATLAB R2016b,

TABLE 1 Morphological, texture and vascular parameters analysed in the study

Morphological features (8 parameters)	General • Area • Perimeter • Circularity Normalized radial length (NRL) • Area ratio (AR) • Standard deviation of NRL (DNRL) • Roughness index (R) Convex polygon • Overlap ratio (OR) • Normalized residual mean square value (nrsv)
Texture features (122 parameters)	First-Order Statistics (FOS) • Mean value • Standard deviation • Skewness • Kurtosis • Median • Mode Gray Level Co-occurrence Matrix (GLCM) • Contrast (0°, 45°, 90°, 135°) • Correlation (0°, 45°, 90°, 135°) • Energy (0°, 45°, 90°, 135°) • Homogeneity (0°, 45°, 90°, 135°) Gray Level Run Length Matrix (GLRLM) • 11 indexes repeated for 0°, 45°, 90°, 135° Gray Level Difference Matrix (GLDM) • Contrast • Angular second moment • Entropy • Mean Fractal Dimension Wavelet Decomposition • Approximation and 3 details for 3 levels Neighborhood Gray Tone Difference Matrix (NGTDM) • Coarseness • Contrast • Busyness • Complexity • Strength
Vascular features (2 parameters)	• Vascularization inside the tumor (vasc_in) • Vascularization outside the tumor • (vasc_out)

MathWorks Inc.). The fivefold cross-validation method was employed to estimate the classification performance of each model in correctly classifying data. For the classification, 23 different classifiers belonging to 6 groups (decision trees, discriminant analysis, logistic regression, support vector machines, nearest neighbour classifiers and ensemble classifiers) were tested.¹⁸

4.1.5 | Statistical analysis

The performance of each classification approach was evaluated calculating sensitivity, specificity and classification accuracy. For each

of the classifiers tested, receiver operating characteristic (ROC) curve was obtained and the area under the ROC curve (AUC) was calculated.

4.2 | RESULTS

Melanocytic naevi and MM appear as hypoechoic fusiform (84% of MN; 95% of MM) or oval (16% of MN, 5% of MM) inhomogeneous lesions, with a variable degree of intralesional vascularization most frequently found in MM instead of MN (85% of MM; 26% of MN) (Figure 1).

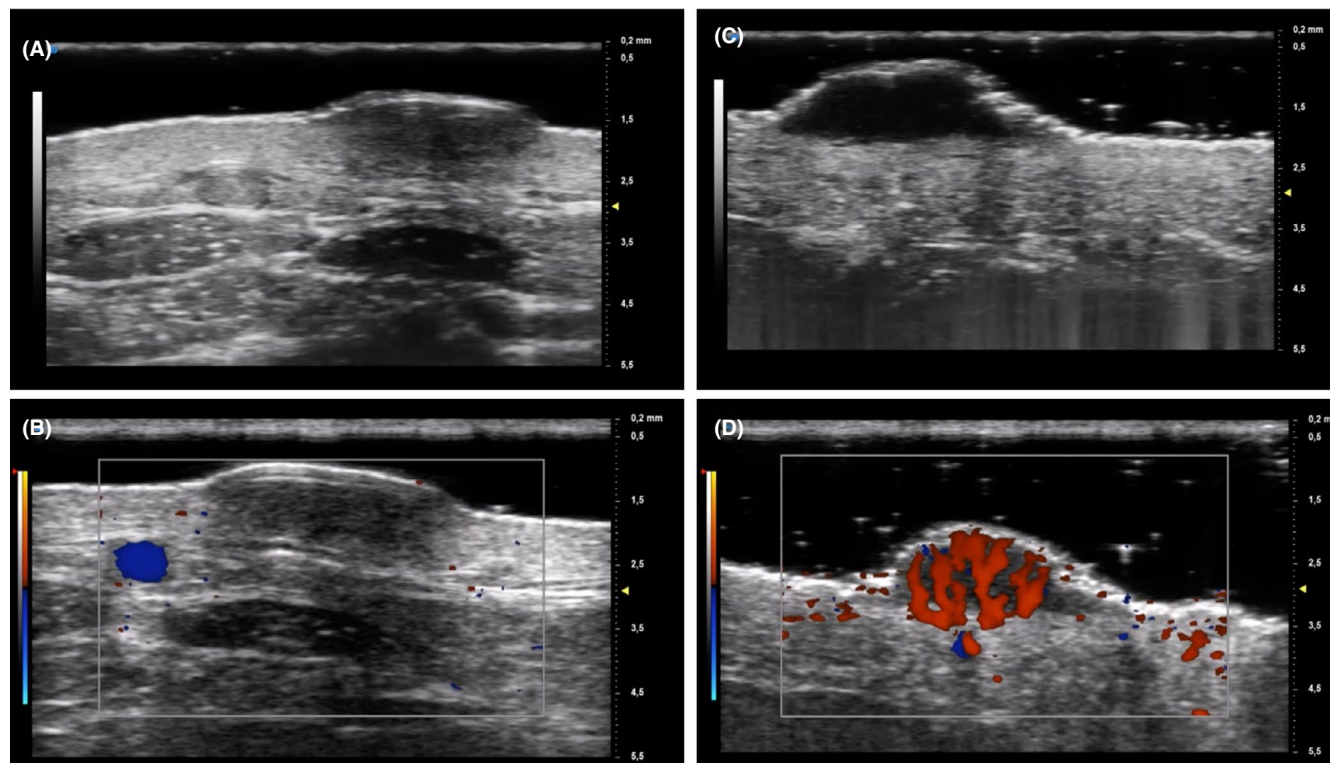


FIGURE 1 Examples of B-mode and colour Doppler images obtained for a melanocytic naevus (A, B) and a malignant melanoma (C, D), respectively. The melanocytic naevus appears as an oval inhomogeneous non-vascularized hypoechoic lesion (A, B); instead, the melanoma appears as a fusiform vascularized inhomogeneous hypoechoic lesion (C, D)

Principal component analysis applied on the complete set of the 132 features yielded to a total of 36 principal components of which only the first one was employed for testing the different classifiers. The first PCA explains >95% of the data variance. The classifiers providing the best sensitivity value (100%) turned out to be the coarse KNN (k-Nearest Neighbour) and the boosted trees algorithms, while simple tree, weighted KNN and bagged trees approaches provided the higher specificity value (70%). The best accuracy (76.9%) was obtained with the weighted KNN and bagged trees approaches and the formed one provided also the the higher AUC value (83%). The classifier leading to the best combination of the performance parameters calculated is the weighted KNN classifies; this combination led to accuracy of 76.9%, AUC of 83%, sensitivity of 84% and specificity of 70%.

5 | RELEVANCE AND PERSPECTIVES

To the best of our knowledge, this paper is the first to employ UHFUS to acquire and compare images of MM and MN. In our study, the intralesional vascularization as isolated parameter seems to be not sufficient in differentiating MM from MN. In general, our best classifier performed very well in terms of accuracy and AUC of the ROC curve. The described system based on UHFUS and a machine learning approach seems to be an effective and non-invasive diagnostic tool for the differential diagnosis of MN and MM.

One of the main advantages of our approach is that a full set of different classification algorithms was tested thus finding the classifier that offers best performances according to suitable metrics. Furthermore, physicians, looking at the full set of classifiers, could select the one which maximizes the most appropriate parameters leading to an optimal decision for any specific clinical setting. Another key aspect of our approach is that it includes the evaluation of a conspicuous number of ultrasonographic features belonging to different domains with texture, morphological and functional parameters mixed up. In this work, machine learning techniques were used to select and classify biomarkers obtained through image processing algorithms applied to regions of interest of the lesion detected by manually identification of lesion borders. However, in a future version of the work, it will be possible to integrate a fully automatic segmentation stage of lesion borders obtained through artificial intelligence algorithms, such as those proposed by Costa et al.¹⁹

The UHFUS allows clinicians to perform an accurate preoperative evaluation of skin and mucosal lesions^{20–23} and may potentially assist the dermatological approach on melanocytic lesions, but further studies with a large sample size are needed to validate the protocol. In particular, the use of a machine learning approach in processing ultrasonographic images could be useful for clinicians, amplifying the possibility to differentiate between benign and malign lesions.

ACKNOWLEDGEMENTS

This article has no funding source.

CONFLICT OF INTERESTS

None.

AUTHOR CONTRIBUTIONS

TO, SV and VD performed the ultrasound acquisitions. FF, NDL, TO, VD and FC carried out the study. TO and NDL wrote the manuscript with support from FF, VD and MR. GA and AJ contributed to the interpretation of the results. All authors discussed the results and contributed to the final manuscript.

ORCID

Teresa Oranges  <https://orcid.org/0000-0002-7277-3278>

REFERENCES

- Apalla Z, Lallas A, Sotiriou E, Lazaridou E, Ioannides D. Epidemiological trends in skin cancer. *Dermatol Pract Concept*. 2017;7:1-6.
- Balch CM, Gershenwald JE, Soong SJ, et al. Final version of 2009 AJCC melanoma staging and classification. *J Clin Oncol*. 2009;27:6199-6206.
- Benelli C, Roscetti E, Pozzo VD, Gasparini G, Cavicchini S. The dermoscopic versus the clinical diagnosis of melanoma. *Eur J Dermatol*. 1999;9:470-476.
- Bono A, Bartoli C, Cascinelli N, et al. Melanoma detection. A prospective study comparing diagnosis with the naked eye, dermoscopy and telespectrophotometry. *Dermatology*. 2002;205:362-366.
- Zalaudek I, Argenziano G, Soyer HP, et al. Three-point checklist of dermoscopy: an open internet study. *Br J Dermatol*. 2006;154:431-437.
- De Giorgi V, Savarese I, Rossari S, et al. Features of small melanocytic lesions: does small mean benign? A clinical-dermoscopic study. *Melanoma Res*. 2012;22:252-256.
- Kittler H, Pehamberger H, Wolff K, Binder M. Diagnostic accuracy of dermoscopy. *Lancet Oncol*. 2002;3:159-165.
- Flóristán Muruzábal U, Gamo Villegas R, Pampín Franco A, Pinedo Moraleda F, Pérez Fernández E, López-Esteban JL. Combined in vivo reflectance confocal microscopy and digital dermoscopy for follow up of patients at high risk of malignant melanoma: a prospective case series study. *J Dermatol*. 2017;44:681-689.
- Catalano O, Caracò C, Mozzillo N, Siani A. Locoregional spread of cutaneous melanoma: sonography findings. *Am J Roentgenol*. 2010;194:735-745.
- Serrone L, Solivetti FM, Thorel MF, Eibenschutz L, Donati P, Catricalà C. High frequency ultrasound in the preoperative staging of primary melanoma: a statistical analysis. *Melanoma Res*. 2002;12:287-290.
- Wortsman X. Sonography of the primary cutaneous melanoma: a review. *Radiol Res Pract*. 2012;2012:814396.
- Andrėkutė K, Linkevičiūtė G, Raišutis R, Valiukevičienė S, Makštienė J. Automatic differential diagnosis of melanocytic skin tumors using ultrasound data. *Ultrasound Med Biol*. 2016;42:2834-2843.
- Samimi M, Perrinaud A, Naouri M, et al. High-resolution ultrasonography assists the differential diagnosis of blue naevi and cutaneous metastases of melanoma. *Br J Dermatol*. 2010;163:550-556.
- Erickson BJ, Korfiatis P, Akkus Z, Kline TL. Machine learning for medical imaging. *Radiographics*. 2017;37:505-515.
- Wang S, Summers RM. Machine learning and radiology. *Med Image Anal*. 2012;16:933-951.
- Ethun CG, Delman KA. The importance of surgical margins in melanoma. *J Surg Oncol*. 2016;113:339-345.
- Abdi H, Williams LJ. Principal component analysis. In: Gentle JE, Scott DW, eds. *WIREs Comp Stat*. Wiley, New York, NY; 2010:433-459.
- Amancio DR, Comin CH, Casanova D, et al. A systematic comparison of supervised classifiers. *PLoS One*. 2014;9:e94137.
- Costa MGF, Campos JPM, de Aquino E, Aquino G, de Albuquerque Pereira WC, Costa Filho CFF. Evaluating the performance of convolutional neural networks with direct acyclic graph architectures in automatic segmentation of breast lesion in US images. *BMC Med Imaging*. 2019;19:85.
- Izzetti R, Vitali S, Aringhieri G, et al. Ultra-high frequency ultrasound, a promising diagnostic technique: review of the literature and single-center experience. *Can Assoc Radiol J*. 2020;846537120940684. [10.1177/0846537120940684](https://doi.org/10.1177/0846537120940684). <https://doi.org/10.1177/0846537120940684>. [Online ahead of print].
- Oranges T, Vitali S, Benincasa B, et al. Advanced evaluation of hidradenitis suppurativa with ultra-high frequency ultrasound: a promising tool for the diagnosis and monitoring of disease progression. *Skin Res Technol*. 2020;26:513-519.
- Oranges T, Janowska A, Vitali S, et al. Dermatoscopic and ultra-high frequency ultrasound evaluation in cutaneous postradiation angiosarcoma. *J Eur Acad Dermatol Venereol*. 2020;34:e741.
- Izzetti R, Vitali S, Aringhieri G, et al. The efficacy of Ultra-High Frequency Ultrasonography in the diagnosis of intraoral lesions. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2020;129:401-410.

How to cite this article: Fata F, Oranges T, Di Lascio N, et al. Ultra-high-frequency ultrasound and machine learning approaches for the differential diagnosis of melanocytic lesions. *Exp Dermatol*. 2022;31:94-98. <https://doi.org/10.1111/exd.14330>